

## The Program in Action

When you run *simplesubcipher.py*, the output will look like this:

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```
Simple Substitution Cipher, by Al Sweigart
A simple substitution cipher has a one-to-one translation for each
symbol in the plaintext and each symbol in the ciphertext.
Do you want to (e)ncrypt or (d)ecrypt?
> e
Please specify the key to use.
Or enter RANDOM to have one generated for you.
> random
The key is WNOMTRCEHDXBFVSLKAGZIPYJQU. KEEP THIS SECRET!
Enter the message to encrypt.
> Meet me by the rose bushes tonight.
The encrypted message is:
Fttz ft nq zet asgt nigetg zsvhcez.
Full encrypted text copied to clipboard.
```

```
Simple Substitution Cipher, by Al Sweigart
A simple substitution cipher has a one-to-one translation for each
symbol in the plaintext and each symbol in the ciphertext.
Do you want to (e)ncrypt or (d)ecrypt?
> d
Please specify the key to use.
> WNOMTRCEHDXBFVSLKAGZIPYJQU
Enter the message to decrypt.
> Fttz ft nq zet asgt nigetg zsvhcez.
The decrypted message is:
Meet me by the rose bushes tonight.
Full decrypted text copied to clipboard.
```

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## How It Works

The position of each of the key's 26 letters corresponds to the letter of the alphabet at that same position:

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ |
| W | N | O | M | T | R | C | E | H | D | X | B | F | V | S | L | K | A | G | Z | I | P | Y | J | Q | U |

*Figure 66-1: How the letters of the alphabet encrypt with a key that begins with WNOM. To decrypt, replace letters at the bottom with the corresponding letters above them.*

With this key, the letter *A* encrypts to *W* (and *W* decrypts to *A*), the letter *B* encrypts to *N*, and so on. The `LETTERS` and `key` variables are assigned to `charsA` and `charsB` (or the other way around if decrypting). Any message characters in `charsA` are substituted with the corresponding character in `charsB` to produce the final translated message.

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1. `""Simple Substitution Cipher, by Al Sweigart al@inventwithpython.com`
  2. A simple substitution cipher has a one-to-one translation for each
  3. symbol in the plaintext and each symbol in the ciphertext.